RESEARCH & SOFTWARE ENGINEERING

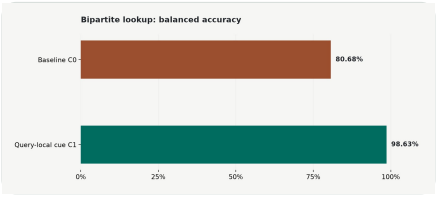
Mehmet Selman Yılmaz

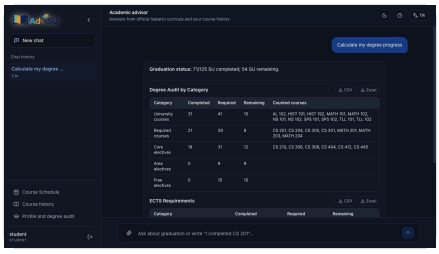
Computer Science and Engineering · Sabancı University

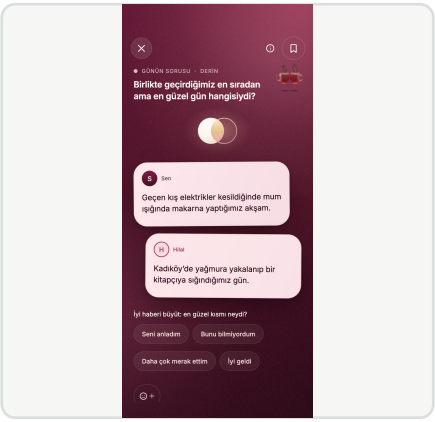
Graduated June 2026 · GPA 3.44 / 4.00

Selected projects

Graph reasoning at KTH. Academic advising with adviSU. Shared mobile experiences with Atcha. Research and software from my internships and university projects.







KTH

adviSU

Atcha

Portfolio website

GitHub

LinkedIn

+90 537 408 10 74 · msyilmaz@kth.se

PROJECT INDEX

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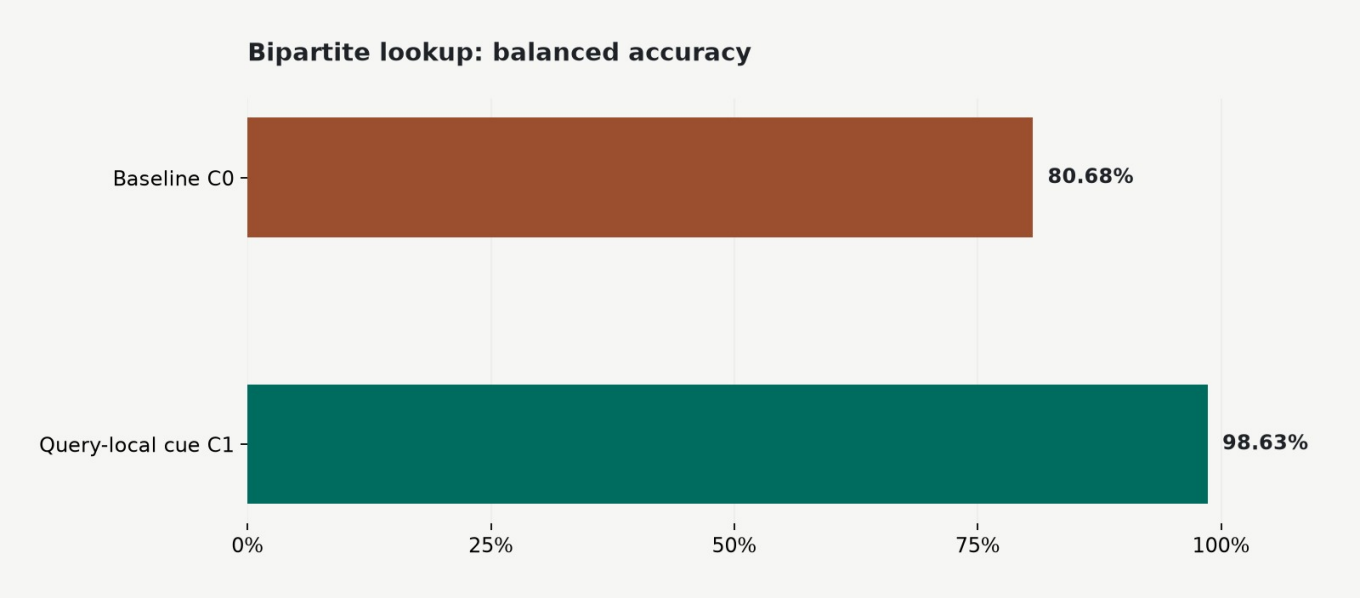
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Education, internships and skills

RESEARCH INTERNSHIP · EXPERIMENT 1

KTH - LLM Graph Reasoning

KTH Royal Institute of Technology · Prof. Aristides Gionis | Jul - Sep 2026

Research on how text encodings affect Llama-3.1-8B graph reasoning.



Bipartite lookup · 120 graphs · 1,750 questions per condition

80.68%
baseline balanced accuracy

98.63%
query-local cue

+17.95 pp
same-question improvement

FEATURES

- Placed a local structural cue near the query and compared it with the baseline encoding.
- Used identical questions and separate scores for same-side, edge and non-edge cases.

RECORDED RESULT

The gain is specific to this bipartite lookup experiment.

Python · PyTorch · Llama-3.1-8B · NetworkX · Slurm · LLM Evaluation

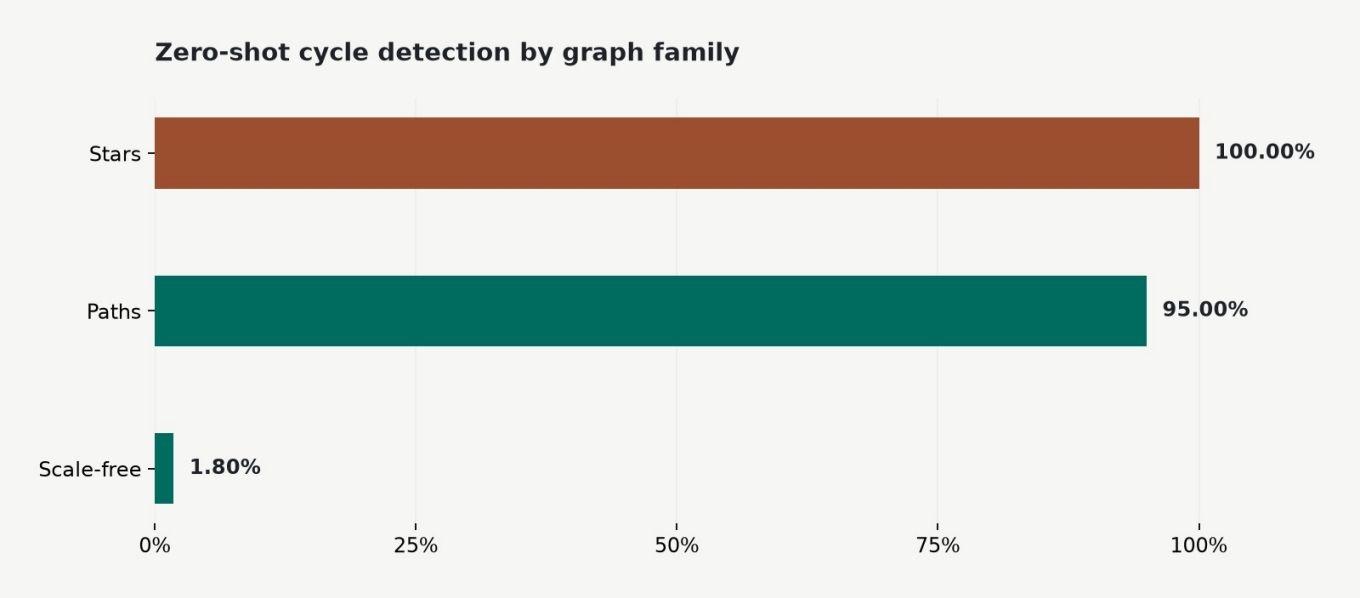
Portfolio website

RESEARCH INTERNSHIP · EXPERIMENT 2

KTH - LLM Graph Reasoning

KTH Royal Institute of Technology · Prof. Aristides Gionis | Jul - Sep 2026

Research on how text encodings affect Llama-3.1-8B graph reasoning.



Zero-shot cycle detection · cells from the recorded graph-structure run

100%
star graphs

95%
path graphs

1.8%
scale-free graphs

FEATURES

• Evaluated 27,600 responses across seven graph families and six tasks.

• Saved raw responses and scoring artifacts for reproduction. Llama-3.1-8B replaces the original paper’s PaLM models.

RECORDED RESULT

Cycle-detection performance varies sharply with graph structure.

Python · PyTorch · Llama-3.1-8B · NetworkX · Slurm · LLM Evaluation

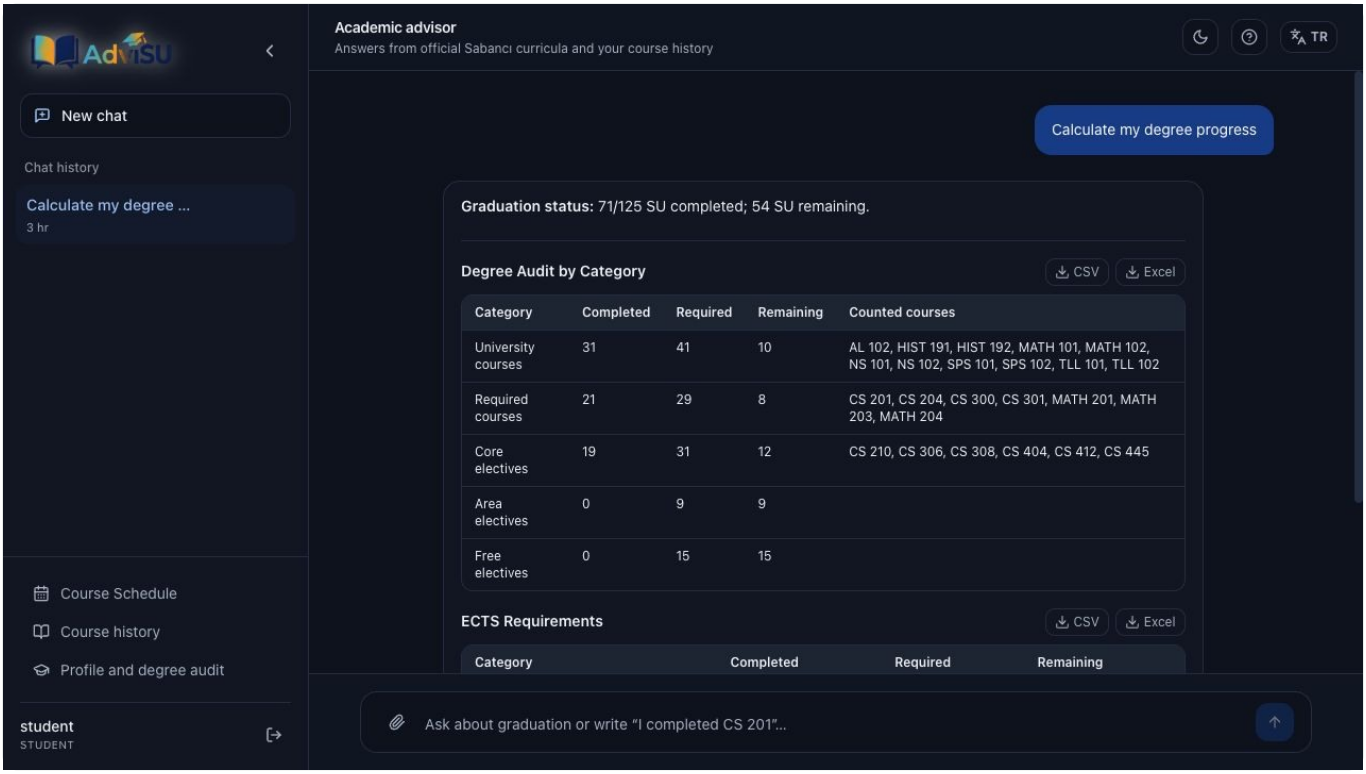
Portfolio website

ACADEMIC ADVISING · CHAT

adviSU

Independent project · Sabancı University curriculum | Apr 2026 - Present

An academic adviser with chat, degree audits and weekly course planning.



Real application UI · official curricula · sample student and grades

FEATURES

• BERT intent detection routes chat requests to retrieval, degree checks or planning.

• Degree audits show counted courses, credits and prerequisite eligibility. Missing evidence prompts clarification.

Python · FastAPI · React · TypeScript · MongoDB · BM25F · E5 Embeddings

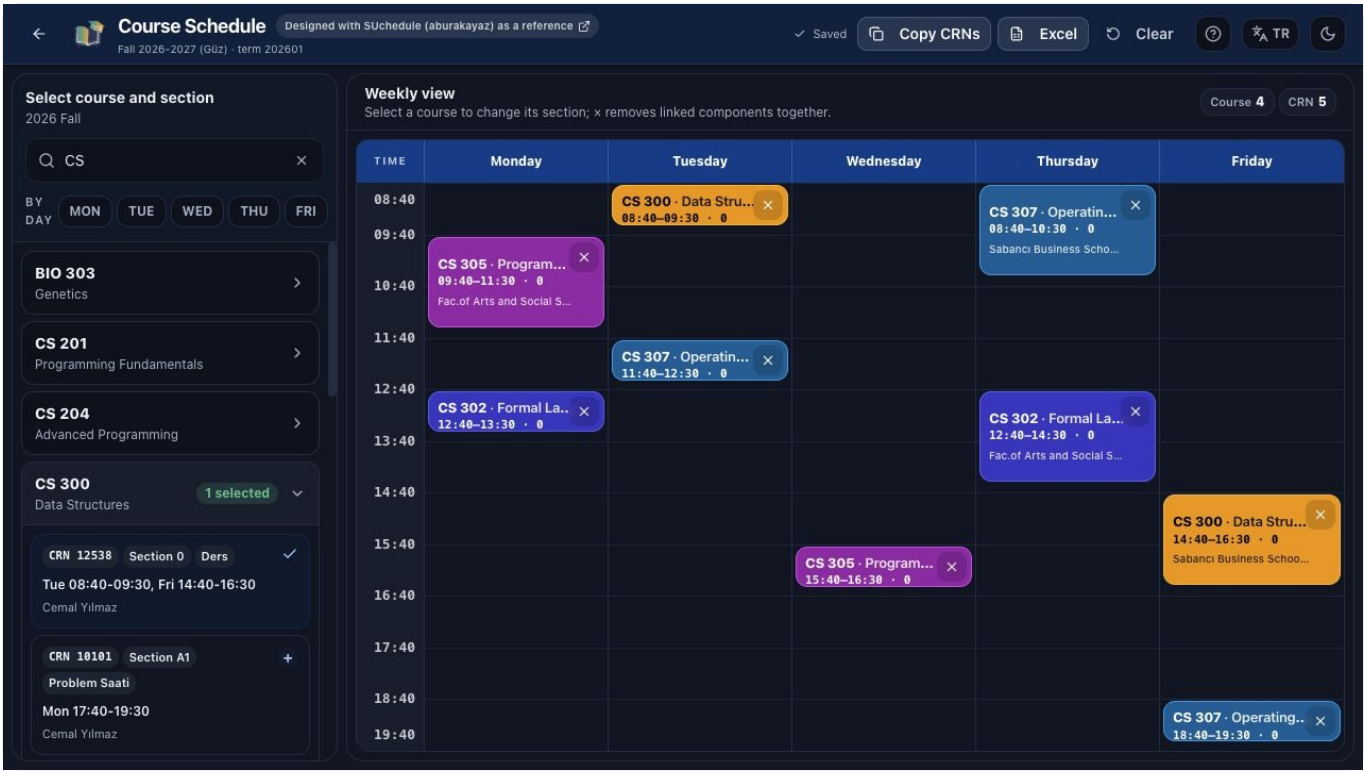
Portfolio website

ACADEMIC ADVISING · SCHEDULE

adviSU

Independent project · Sabancı University curriculum | Apr 2026 - Present

An academic adviser with chat, degree audits and weekly course planning.



Official section data · illustrative course selection

FEATURES

• Backtracking selects compatible lecture, lab and recitation sections without time conflicts.

• Chat and weekly view share one schedule; revision checks prevent stale overwrites. CRN copying and spreadsheet export support registration.

Python · FastAPI · React · TypeScript · MongoDB · BM25F · E5 Embeddings

Portfolio website

ACADEMIC ADVISING · RETRIEVAL

adviSU

Independent project · Sabancı University curriculum | Apr 2026 - Present

An academic adviser with chat, degree audits and weekly course planning.

59.92%	94.19%	+34.27 pp
BM25 Recall@10	hybrid Recall@10	recorded gain

Retrieve	Validate	Respond
BM25F + E5 + RRF	Degree and section rules	Evidence + explanation

FEATURES

- Combines field-weighted BM25F, multilingual E5 and reciprocal-rank fusion.
- Calculations run in deterministic code; retrieved text is checked before explanation.

RECORDED RESULT

499 frozen queries; 30,343 chunks. The largely template-based benchmark measures retrieval recall, not overall advising accuracy. Complete evidence: 1/16 multi-evidence queries.

Python · FastAPI · React · TypeScript · MongoDB · BM25F · E5 Embeddings

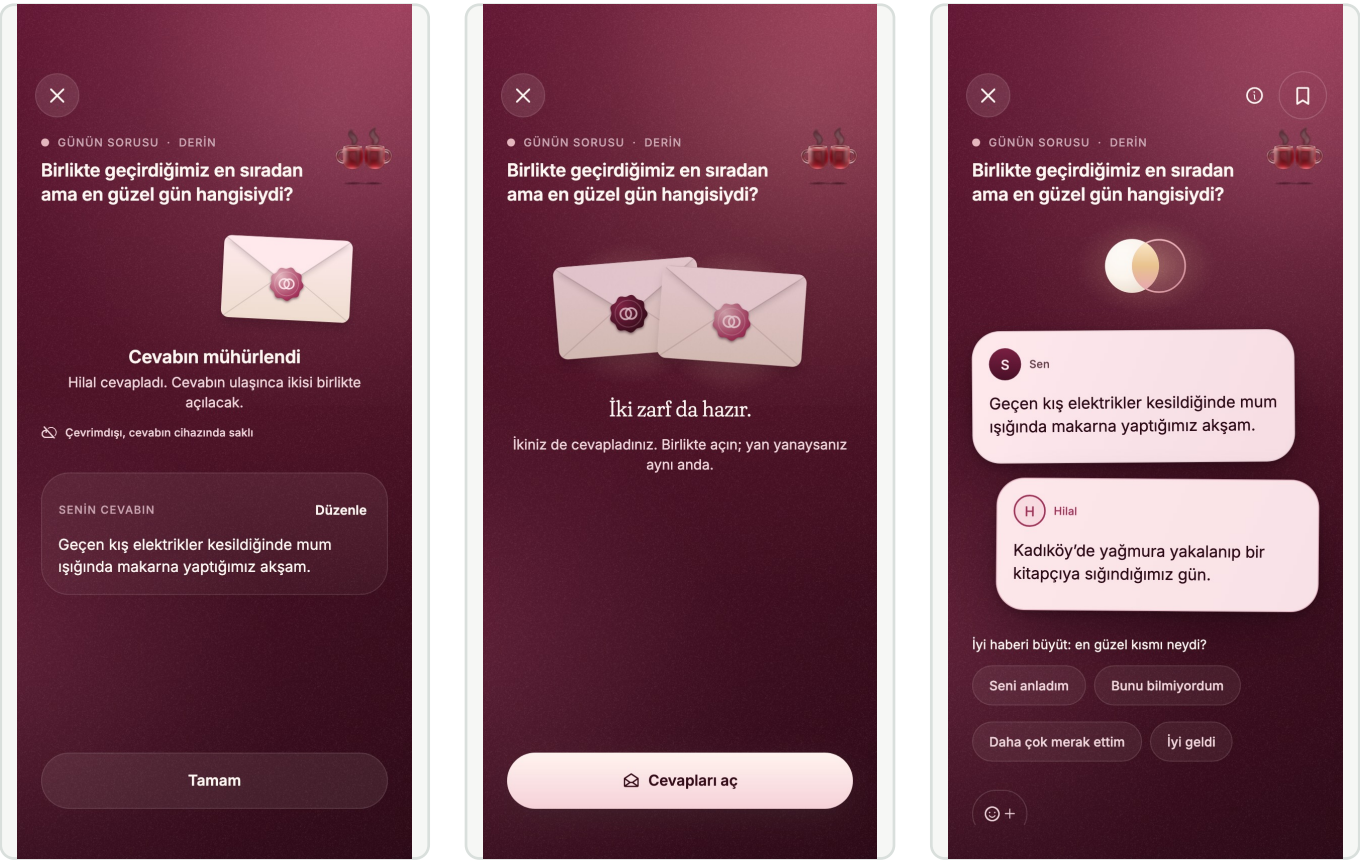
Portfolio website

MOBILE APPLICATION · SHARED QUESTIONS

Atcha

Independent mobile product | Oct 2026 - Present

A two-person mobile app for shared questions, games, chat and memories.



Waiting for partner · Ready to reveal · Revealed answers. Actual visual-test screens; sample content.

FEATURES

• Explicit composing, sending, waiting and reveal states show progress.

• Answers remain locked until both respond. Semantic labels, 44-point targets, reduced motion and 2× text cases support accessibility.

Flutter · Dart · Riverpod · go_router · Supabase · PostgreSQL · Realtime

Portfolio website

MOBILE APPLICATION · SOFTWARE ENGINEERING

Atcha

Independent mobile product | Oct 2026 - Present

A two-person mobile app for shared questions, games, chat and memories.

Client	Server	Database	Realtime
Riverpod state	RPC + RLS checks	Row lock + commit	Signal + refetch

FEATURES

• Server RPCs validate transitions; RLS scopes private content to the authenticated couple.

• Offline retries use backoff and idempotent client IDs. Shared channels have explicit ownership and release.

RECORDED RESULT

Recorded 30-row history query: 500-535 ms -> 180-205 ms; 60/60 result-set checks matched.

Flutter · Dart · Riverpod · go_router · Supabase · PostgreSQL · Realtime

Portfolio website

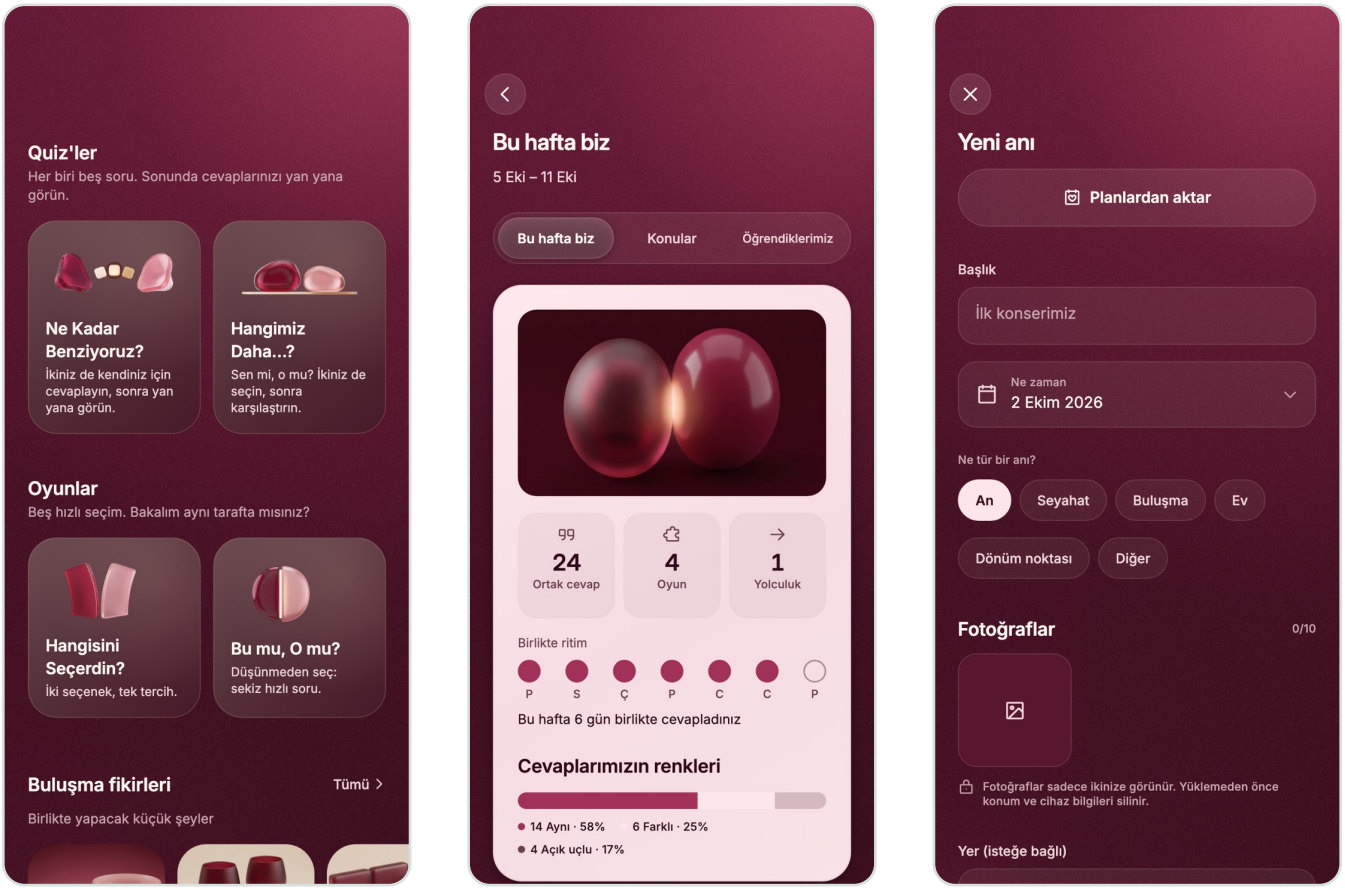
500-535 ms	180-205 ms	60/60
history query before	after; 30 rows	matching result sets

MOBILE APPLICATION · GAMES & MEMORIES

Atcha

Independent mobile product | Oct 2026 - Present

A two-person mobile app for shared questions, games, chat and memories.



Play · Weekly view · Memories. Long test renders shown through a standard phone viewport.

FEATURES

• Shared games, weekly activity and photo memories use the same account and privacy boundaries.

• On-device Sentence-BERT offers conservative answer-similarity hints; negated, truncated and ambiguous inputs abstain.

RECORDED RESULT

Raw answers stay on-device for similarity inference; the score is not a relationship assessment.

Flutter · Dart · Riverpod · go_router · Supabase · PostgreSQL · Realtime

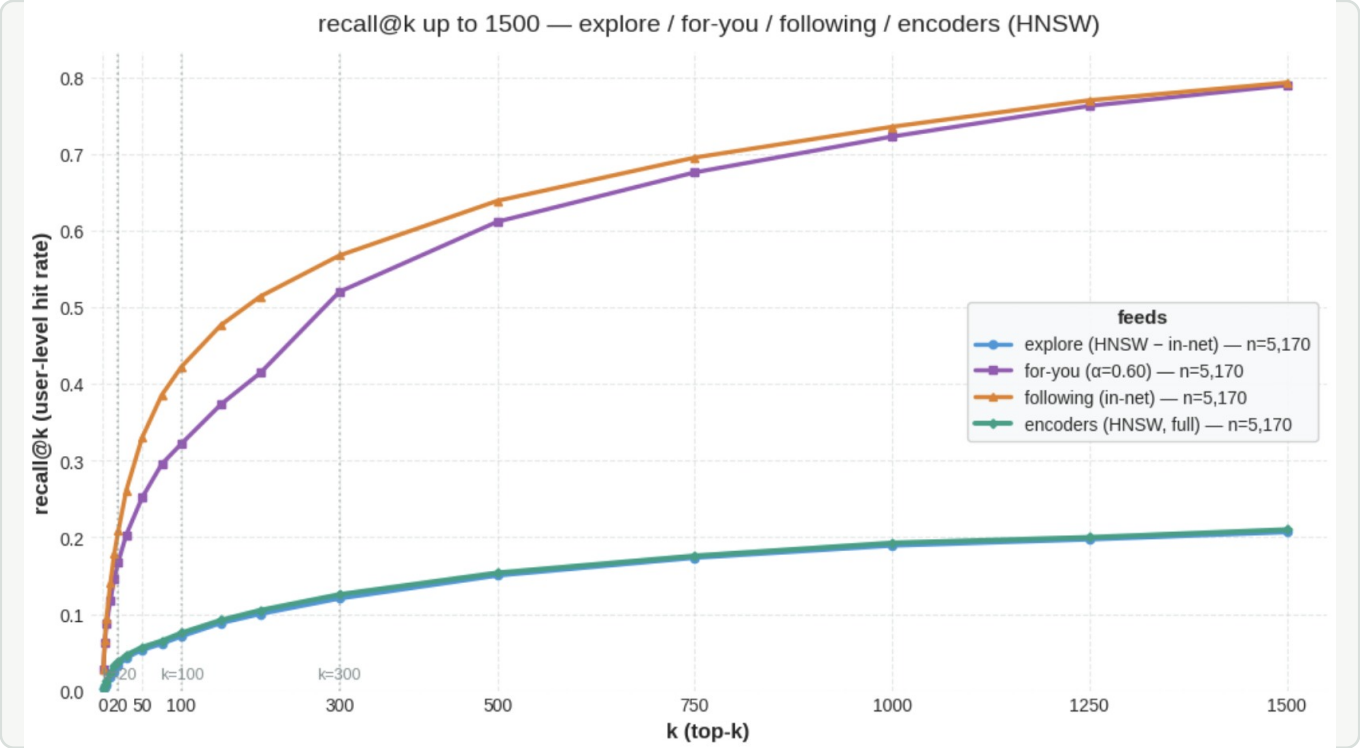
Portfolio website

SELECTED PROJECT

Baykar - Recommendation System

Baykar | Aug - Sep 2025

A social-media recommendation pipeline built during my AI internship.



Modeling & Flow

FEATURES

• Two-Tower retrieval selects candidates; Wide & Deep ranks them.

• TurkishBERTweet and interaction signals cover Instagram, Twitter and Next Sosyal.

RECORDED RESULT

Offline evaluation covered 5,170 users across three platforms.

PyTorch · TurkishBERTweet · Two-Tower · Wide & Deep · Embeddings · Recommendation Systems

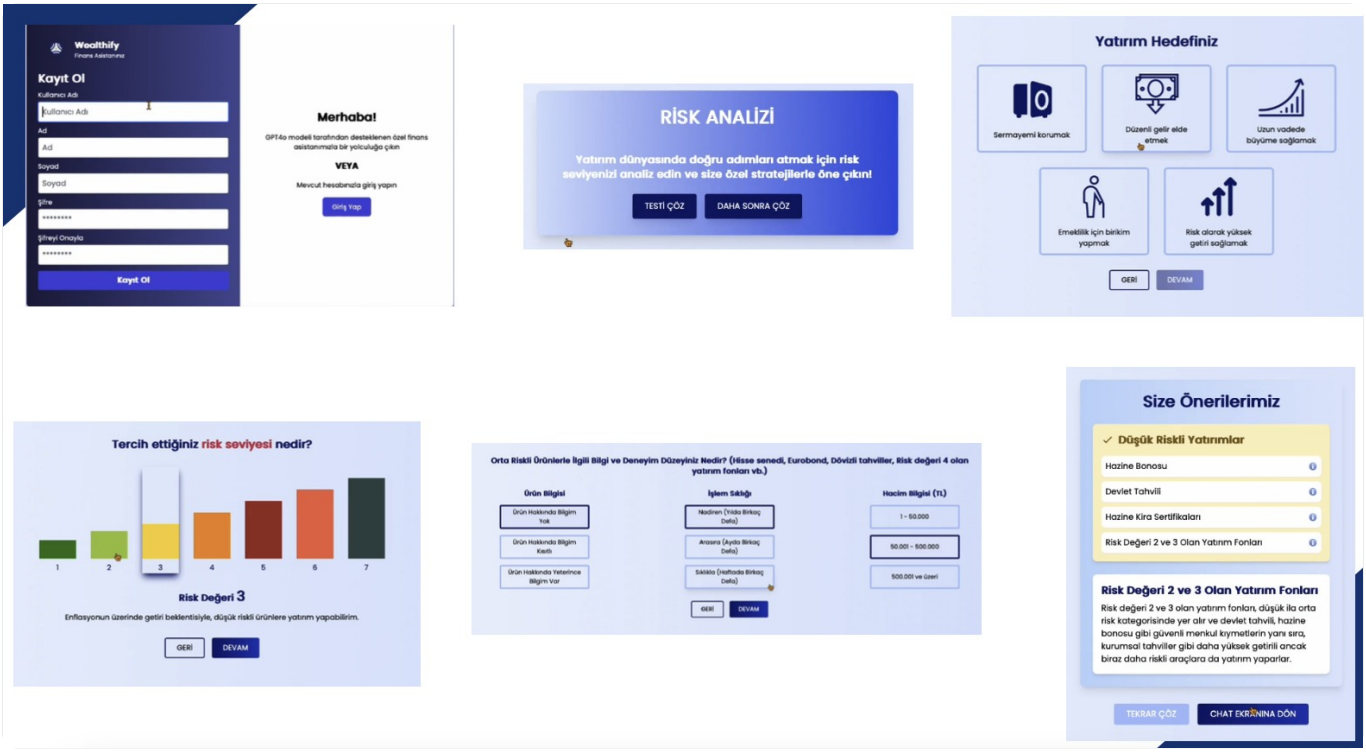
Portfolio website

SELECTED PROJECT

Intertech - Financial RAG

Intertech | Jun - Aug 2024

A financial assistant that answers questions using daily bulletins.



Ingestion & Storage

FEATURES

- Automated bulletin collection, Python processing and Azure Blob storage.
- Node.js/Next.js application with Docker, Kubernetes and Azure CI/CD.

Node.js · Next.js · SQL · Python · RAG · Azure Blob Storage · Web Scraping

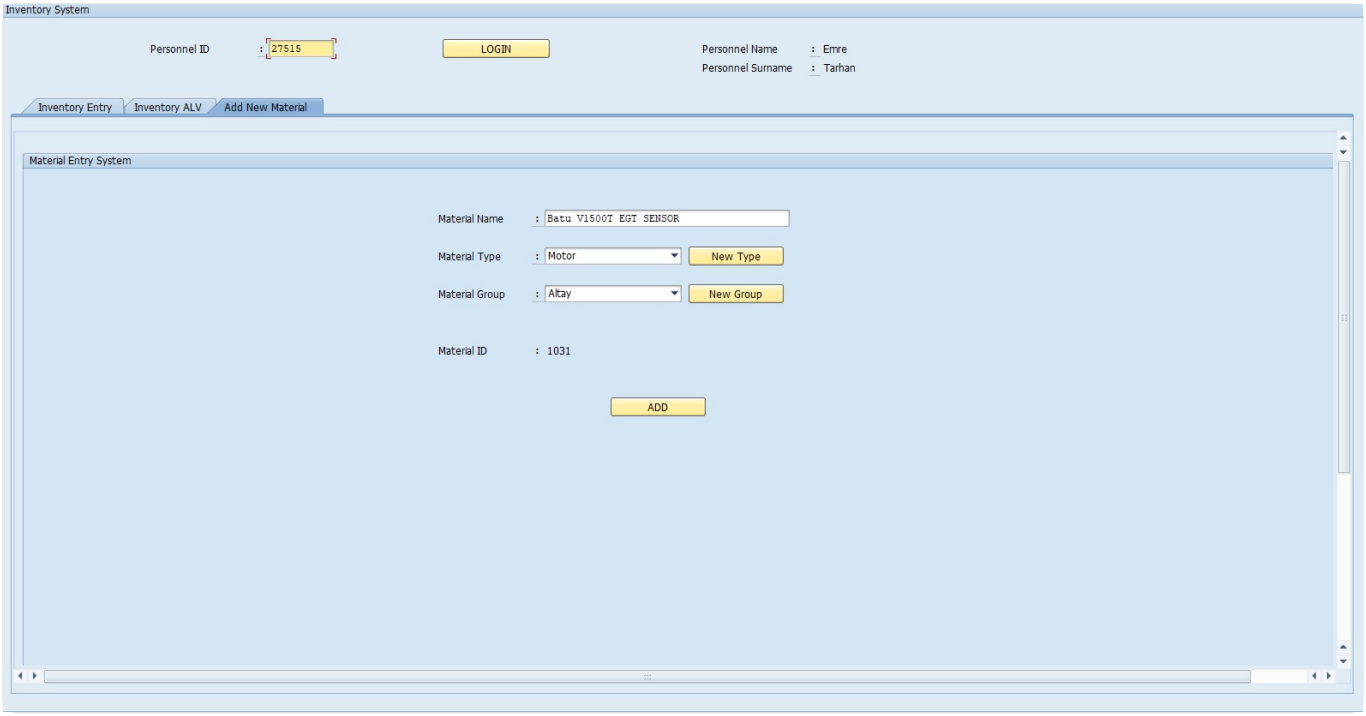
Portfolio website

SELECTED PROJECT

BMC - Inventory Assistant

BMC | Jul - Aug 2025

A natural-language assistant for SAP inventory workflows.



RAG Flow & System

FEATURES

• NestJS/Next.js and OpenAI connect the interface to inventory data.

• ABAP material tracking and automated low-stock alerts.

NestJS · Next.js · OpenAI API · RAG · SAP · ABAP · TypeScript

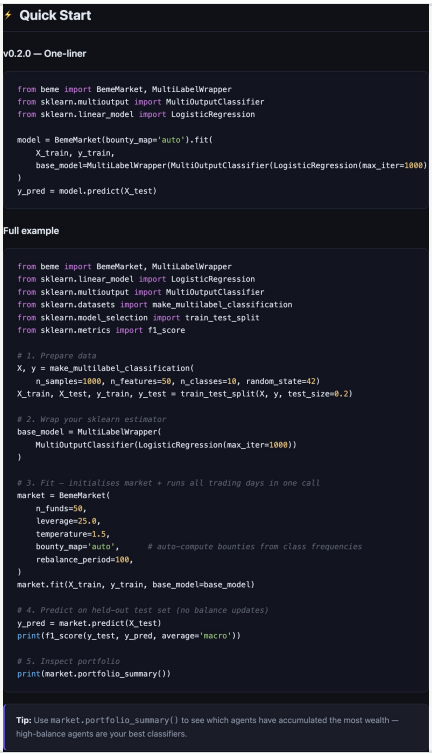
Portfolio website

SELECTED PROJECT

AutoBEME

Open-source / Personal Project | Apr 2026 - Present

A scikit-learn-compatible ensemble for imbalanced classification.



Quick Start

v0.2.0 — One-liner

```
from beme import BemeMarket, MultiLabelWrapper
from sklearn.multioutput import MultiOutputClassifier
from sklearn.linear_model import LogisticRegression

model = BemeMarket(bounty_map='auto').fit(
    X_train, y_train,
    base_model=MultiLabelWrapper(MultiOutputClassifier(LogisticRegression(max_iter=1000)))
)
y_pred = model.predict(X_test)
```

Full example

```
from beme import BemeMarket, MultiLabelWrapper
from sklearn.linear_model import LogisticRegression
from sklearn.multioutput import MultiOutputClassifier
from sklearn.datasets import make_multilabel_classification
from sklearn.model_selection import train_test_split
from sklearn.metrics import f1_score

# 1. Prepare data
X, y = make_multilabel_classification(
    n_samples=1000, n_features=50, n_classes=10, random_state=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# 2. Wrap your sklearn estimator
base_model = MultiLabelWrapper(
    MultiOutputClassifier(LogisticRegression(max_iter=1000))
)

# 3. Fit — initialises market + runs all trading days in one call
market = BemeMarket(
    n_funds=50,
    leverage=25.0,
    temperature=1.5,
    bounty_map='auto',      # auto-compute bounties from class frequencies
    rebalance_period=100,
)
market.fit(X_train, y_train, base_model=base_model)

# 4. Predict on held-out test set (no balance updates)
y_pred = market.predict(X_test)
print(f1_score(y_test, y_pred, average='macro'))

# 5. Inspect portfolio
print(market.portfolio_summary())
```

Tip: Use `market.portfolio_summary()` to see which agents have accumulated the most wealth — high-balance agents are your best classifiers.

Quick Start: Compressing the Full Market Math into One API

FEATURES

• Bootstrap classifiers act as agents with virtual capital.

• Capital-weighted logits, class-frequency rewards and periodic rebalancing determine the ensemble vote.

RECORDED RESULT

Recorded text benchmark: Macro-F1 0.826 -> 0.864; best result in two of three data regimes.

Python · scikit-learn · Multi-label Classification · Prediction Markets · Mathematical Modeling · Classifier Chains

Portfolio website

ENSEMBLE LEARNING · ALGORITHM

AutoBEME

Open-source / Personal Project | Apr 2026 - Present

A scikit-learn-compatible ensemble for imbalanced classification.

Agents	Vote	Reward	Update
Bootstrap samples	Capital-weighted logits	Class-frequency signal	Rebalance capital

FEATURES

- Each classifier starts with virtual capital; correct predictions update its weight.
- Class-frequency rewards emphasize rare classes. Periodic rebalancing changes the consensus.

RECORDED RESULT

Macro-F1 improved from 0.826 to 0.864 on the recorded signal-rich text benchmark; the method did not win every regime.

Python · scikit-learn · Multi-label Classification · Prediction Markets · Mathematical Modeling · Classifier Chains

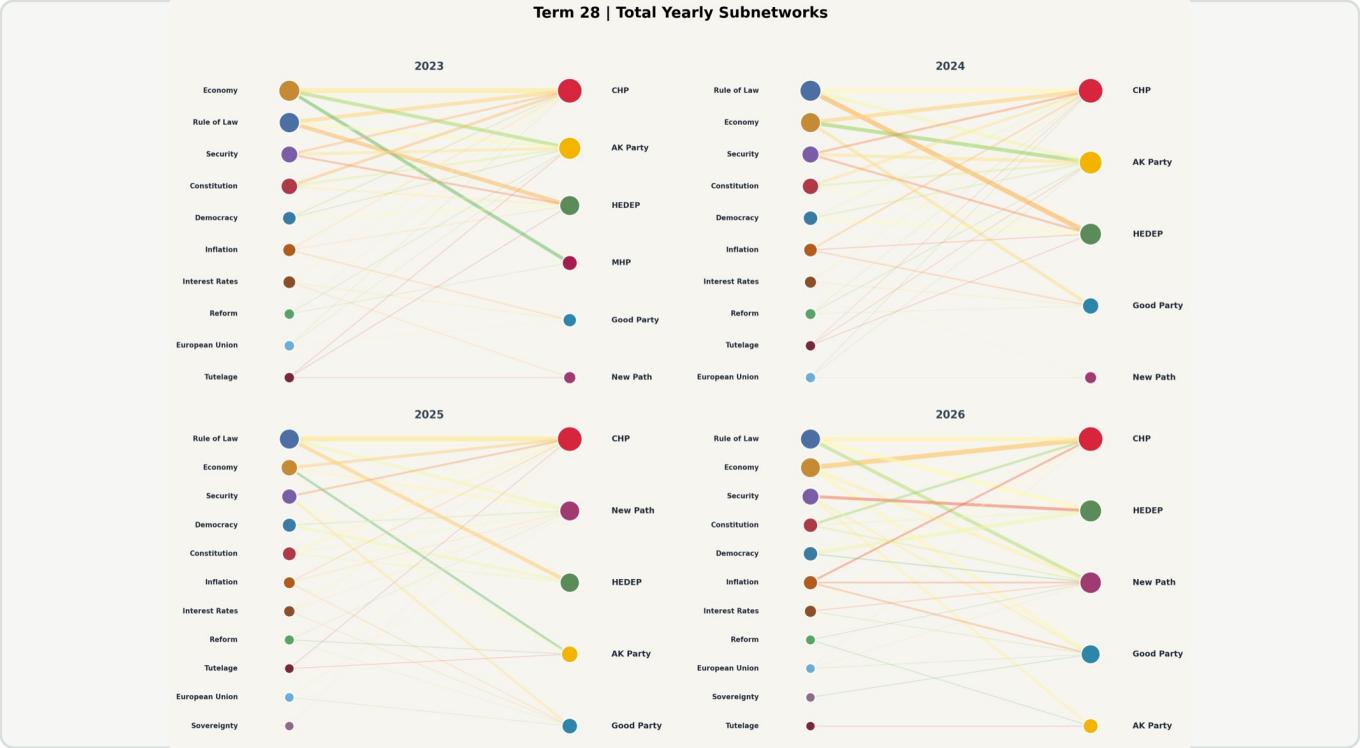
Portfolio website

SELECTED PROJECT

TBMM - Parliamentary Corpus

Sabanci University Research Project | 2025 - Present

A Turkish parliamentary corpus and NLP evaluation pipeline.



Yearly Network Evolution

FEATURES

- Repairs OCR, speaker identities and historical party labels.
- Builds term-specific datasets and compares frozen-transfer BERT baselines with AutoBEME.

RECORDED RESULT

1.33M+ speeches, 2,829 transcripts and 5,017 speakers. Separate TBMM Macro-F1: 0.624 -> 0.640.

Python · Network Science · NLP · Data Cleaning · Political Economy · Research Dashboards

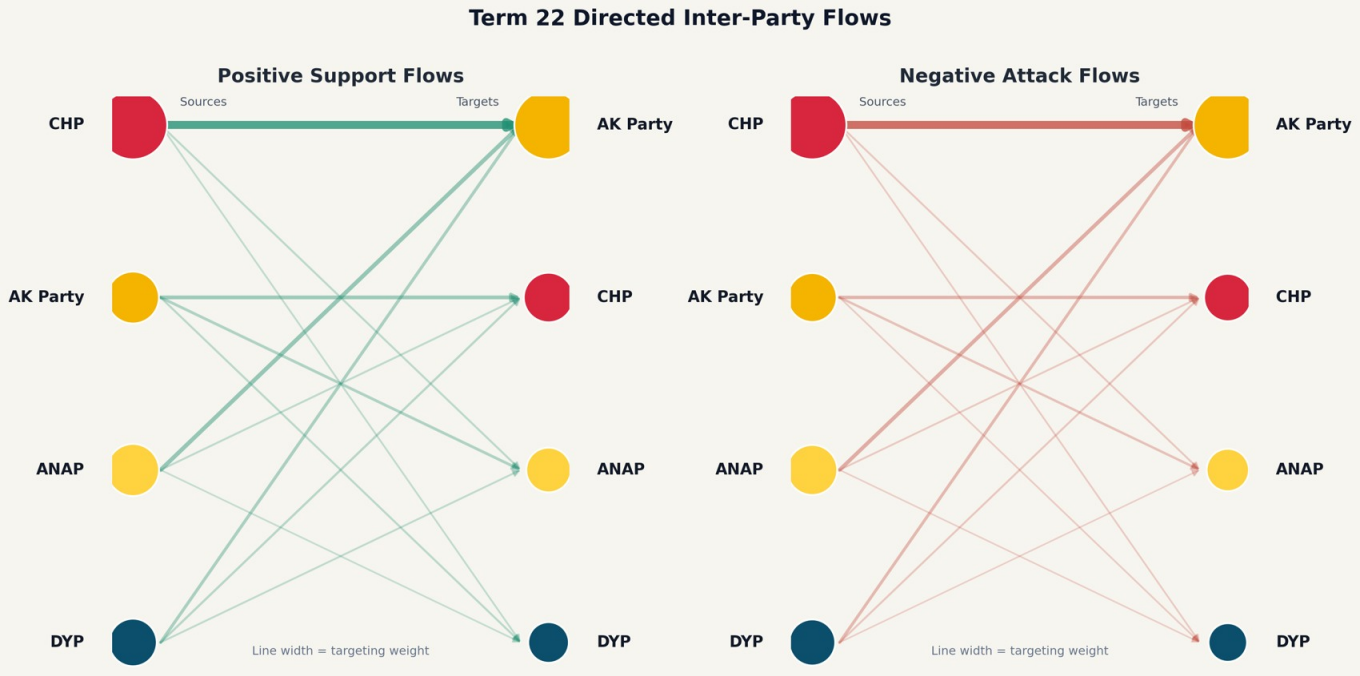
Portfolio website

SELECTED PROJECT

Political-Economy Networks

Academic Research Project | Jan 2026 - Present

Research linking parliamentary language and network structure to economic indicators.



Directed Inter-Party Flows

FEATURES

- Directed and signed networks model speaker and party relationships.
- NetworkX and Statsmodels support centrality, semantic-distance and event-window analysis.

RECORDED RESULT

A 733-row human-annotation pack supports validation; causal conclusions are not established.

Python · NetworkX · Pandas · Statsmodels · Matplotlib · NLP · TBMM

Portfolio website

ALGORITHMS & SEQUENCE MODELS

SparseViz

Research contribution | Sabancı PURE Research Program

A C++ library for sparse matrix and tensor analysis.

6×
recorded ordering speedup

C++
sparse matrix / tensor work

- Implemented ordering algorithms and profiled sparse workloads.
- Compared algorithm behavior within the research library.

Portfolio website

Intent Detection & Slot Filling

Joint intent classification and token-level slot prediction on SNIPS.

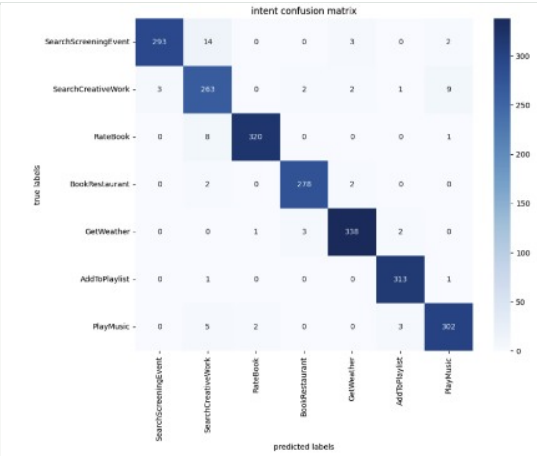


Figure B1: Confusion Matrix BILSTM

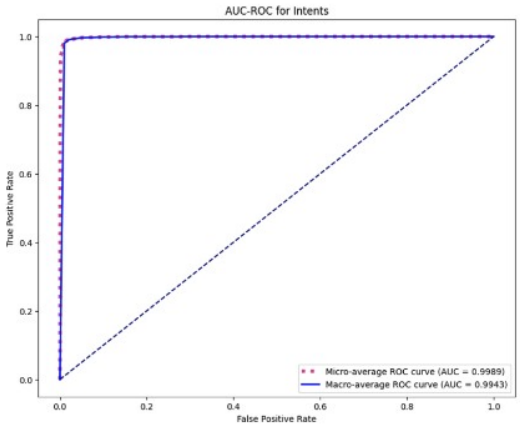


Figure B2: Precision Recall Curve for BILSTM

Existing SNIPS project evaluation

- Slot-Gated BiLSTM with FastText representations.
- Compared LR, XGBoost, LSTM and BiLSTM using Bayesian hyperparameter search.

Reported intent accuracy: 97%; slot accuracy: 79%.

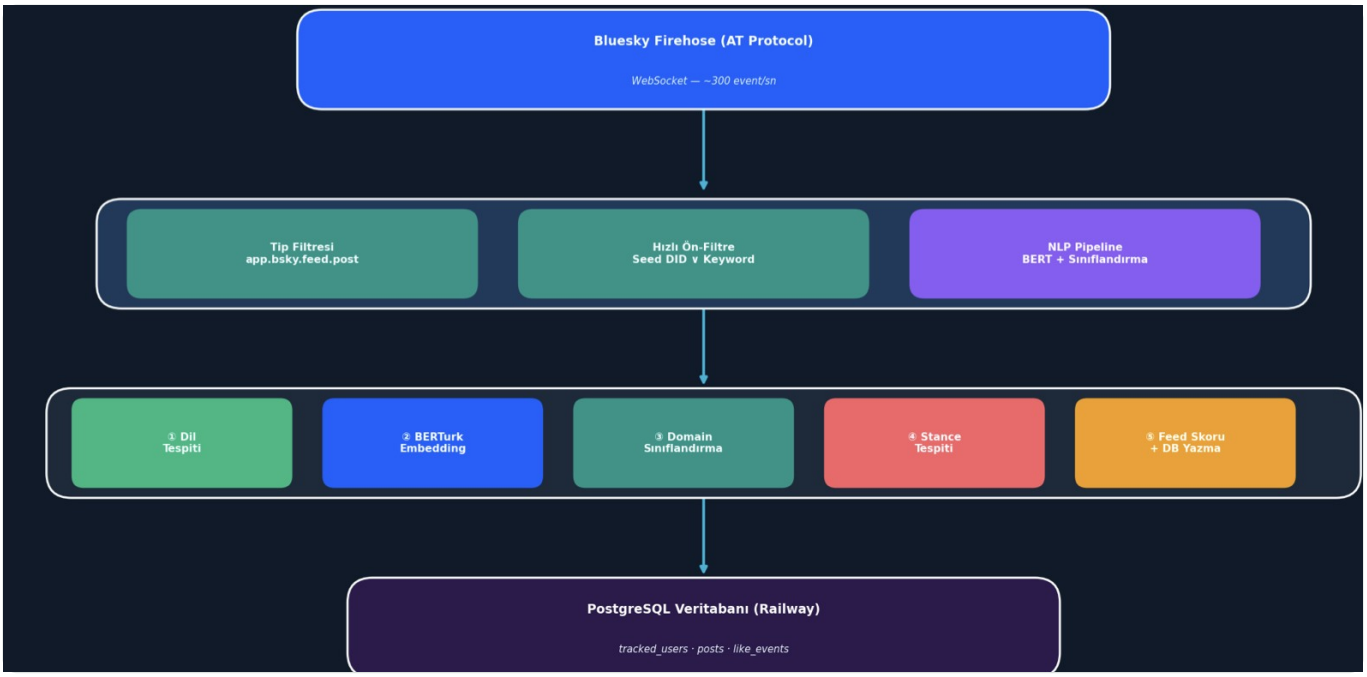
Portfolio website

SELECTED PROJECT

BlueSky - Turkish Topic Feeds

Graduation Project | Graduation project · 2026

A completed graduation project that creates Turkish topic-based BlueSky feeds.



System Architecture

FEATURES

• Filters the AT Protocol firehose, then applies five NLP stages and 768-dimensional BERTurk embeddings.

• Flask serves precomputed PostgreSQL results; Railway hosts the feed service.

RECORDED RESULT

Two feeds, initialized from 270 seed accounts.

AT Protocol · Bluesky Firehose · BERTurk · Flask · PostgreSQL · Railway · WebSocket

Portfolio website

SELECTED PROJECT

Music Emotion Recognition

Deep Learning Project | Sep 2025 - Jan 2026

Predicts arousal and valence from music using the DEAM dataset.

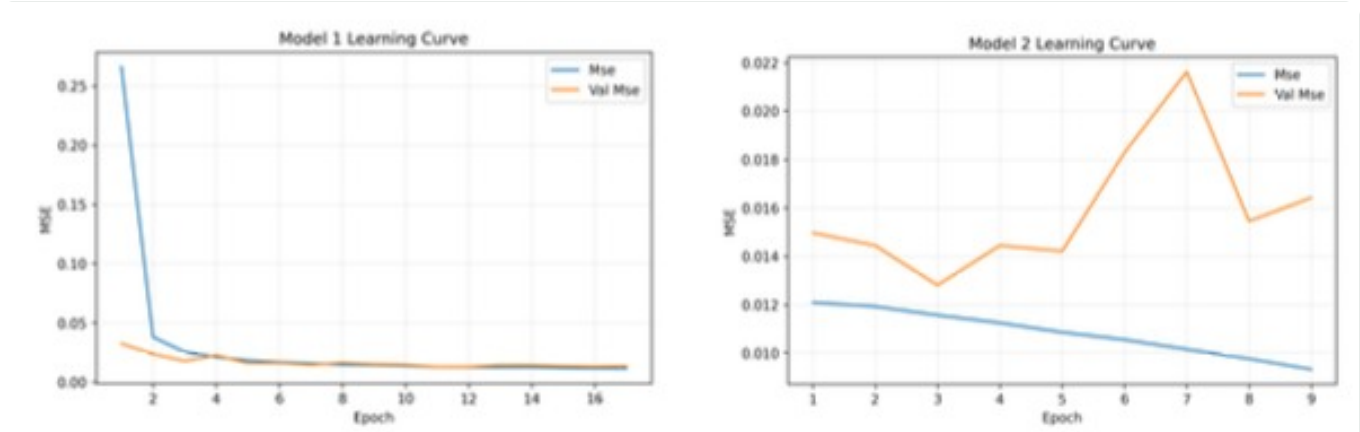


Figure 3. Learning curves (train/val loss over epochs). Model 1 is on the left, and Model 2 is on the right. Model 2 shows faster convergence and more stable training dynamics.

Pipeline & Features

FEATURES

- Five-second log-mel windows with 50% overlap.
- CNN features, attention pooling and a head ensemble; normalization uses training data only.

RECORDED RESULT

Recorded test results: MSE 0.0114; Pearson 0.721; CCC 0.689.

PyTorch · CNN · Attention Pooling · Log-Mel · Regression · DEAM

Portfolio website

SELECTED PROJECT

EchoSpaceAR

HCI Project | Sep 2025 - Present

An AR prototype that turns environmental sounds and speech into visual cues.



AR UI & Cue/HUD Flow

FEATURES

- YAMNet recognizes sound events; Whisper transcribes speech.
- Unity/C# places contextual cues and a HUD in the AR scene.

Unity · C# · AR UI · YAMNet · Whisper · AutoBEME · HCI

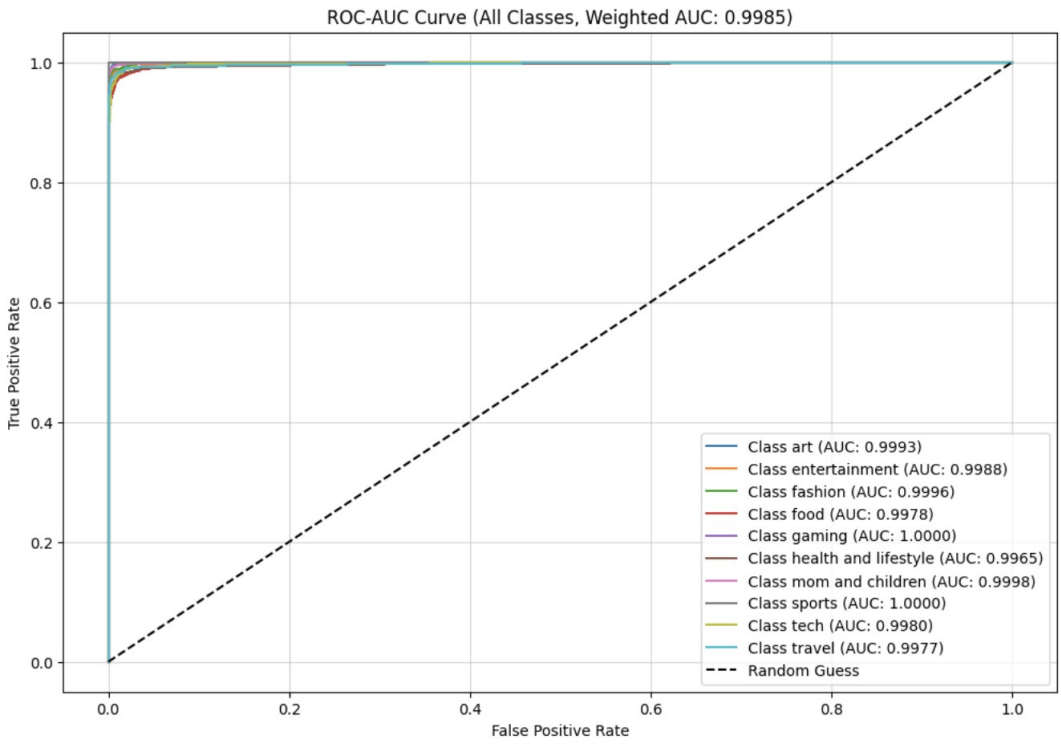
Portfolio website

SELECTED PROJECT

Influencer & Engagement Prediction

Personal Project | Sep 2024 - Jan 2025

Classifies influencer profiles and predicts post likes.



Data & Preprocessing

FEATURES

• Random Forest + Logistic Regression stacking with SMOTE and domain-specific preprocessing.

• A separate Random Forest regressor estimates likes.

RECORDED RESULT

Reported classification F1: 86.7%.

Scikit-learn · Random Forest · Logistic Regression · SMOTE · Stacking · Regression

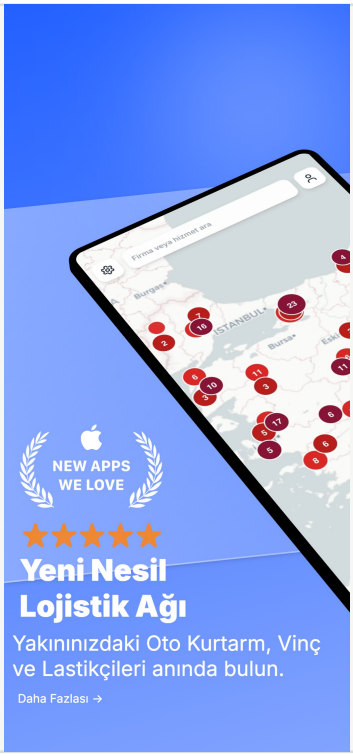
Portfolio website

SELECTED PROJECT

Transport 245

Independent Product · Transport 245 | Feb 2026 - Present

A mobile-ready transport marketplace with maps, orders, ratings and chat.



Map-First Logistics Discovery

FEATURES

• Next.js/React and MapLibre for discovery; NestJS/MongoDB for service data.

• Capacitor provides native packaging; Firebase handles authentication.

RECORDED RESULT

Native packages were prepared; app-store publication is not confirmed.

Next.js 15 · React 19 · Capacitor · NestJS · MongoDB · Firebase Auth · MapLibre

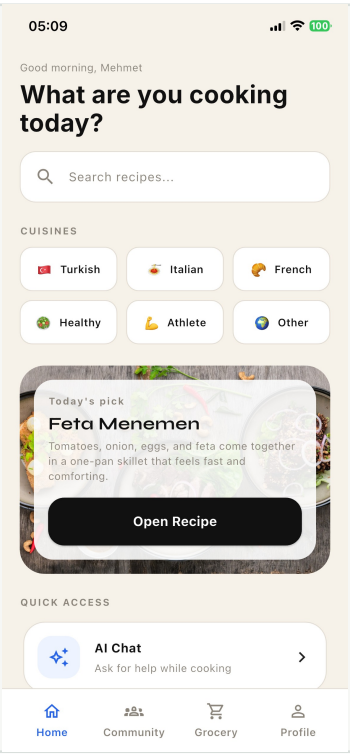
Portfolio website

SELECTED PROJECT

ChefInPocket

Sabancı University · CS310 Team Project | Spring 2026

A recipe app with filtering, account access and step-by-step cooking.



Home and Recipe Discovery

FEATURES

- Flutter client with six Node.js services behind an API gateway.
- Docker/Nginx deployment and unit/widget tests for filtering and login.

Flutter · Node.js · Express · MongoDB · Microservices · Docker · Nginx

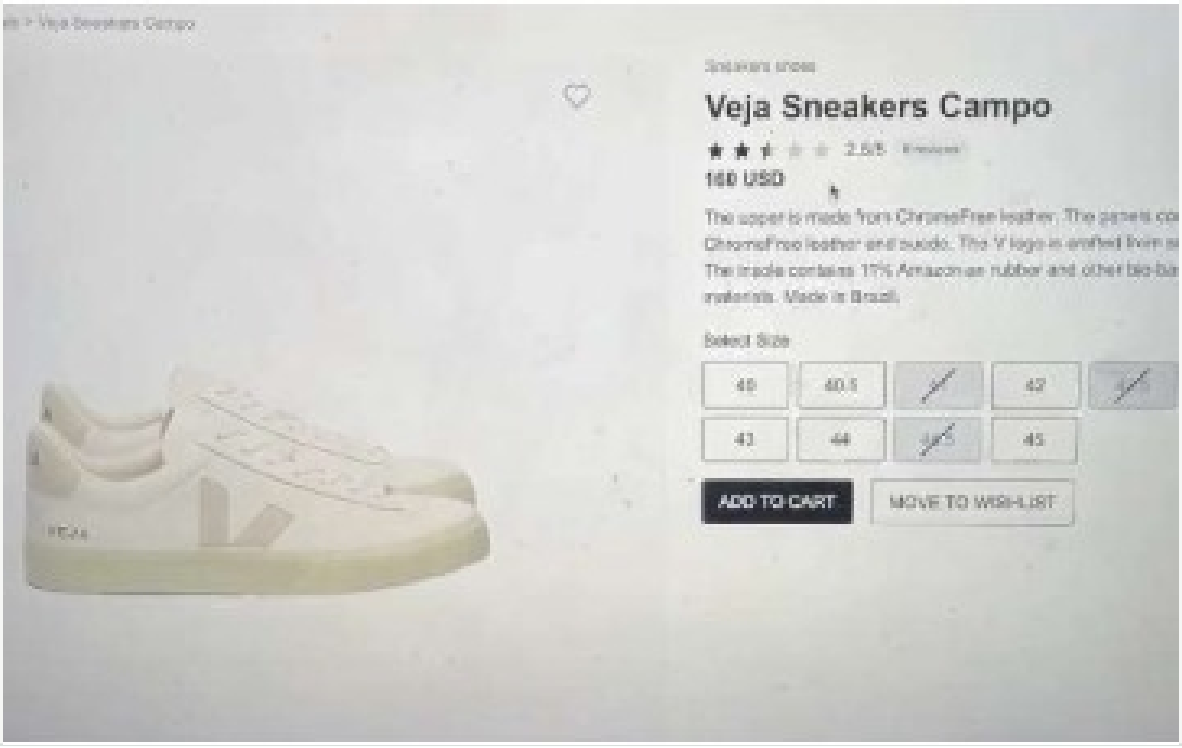
Portfolio website

SELECTED PROJECT

SneakPeek

Personal Project | Mar 2025 - Jun 2025

A sneaker-shopping application with a GPT-based product recommender.



System & AI Recommender

FEATURES

- Eight NestJS services with Kafka/gRPC, JWT and Docker.
- Next.js client; unit, integration and end-to-end tests cover the shopping flows.

NestJS · Next.js · Microservices · OpenAI GPT · JWT · Kafka · Docker

Portfolio website

EDUCATION & EXPERIENCE

Background & contact

Mehmet Selman Yılmaz

EDUCATION

Sabancı University · B.Sc. Computer Science and Engineering
Graduated 24 June 2026 · GPA 3.44 / 4.00

KTH · Jul-Sep 2026

Llama graph reasoning and benchmark reproduction

Baykar · Aug-Sep 2025

Social-media recommendation pipelines

BMC · Jul-Aug 2025

SAP inventory and stock-alert workflows

Intertech · Jun-Aug 2024

Financial RAG and cloud delivery

Sabancı University · Jun 2023-Jun 2024

C++ teaching support for 500+ students

SKILLS IN CONTEXT

Llama / PyTorch / Slurm - KTH
BERT / RAG / FastAPI - adviSU
LSTM / BiLSTM - Intent Detection
C++ - SparseViz
Flutter / PostgreSQL - Atcha
Unity / C# - EchoSpaceAR

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GitHub